SHUBHAM KUMAR

Senior Software Engineer | 7.8+ Years System Design & Backend Architecture

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7.8+ years of expertise architecting and delivering large-scale, service-oriented systems with focus on high performance, reliability, and scalability. Expert in C++ and Python with deep mastery of object-oriented design, distributed systems architecture, and low-latency optimization. Proven track record designing modular, stable systems serving millions of concurrent users. Strong communicator with self-starter mentality and demonstrated ownership across complex technical initiatives and cross-functional teams.

CORE COMPETENCIES

System Architecture & Design: HLD/LLD of large-scale service-oriented systems • Distributed & multithreaded architecture • Object-oriented design principles • Design patterns • Data structures & algorithms • Microservices • API design

Technical Expertise: C++ (Modern C++11/17/20, STL, concurrency) • Python • Backend development • System software • Performance optimization • Caching strategies • Concurrent programming • Memory management

Quality & Testing: CI/CD pipelines • Unit testing & test frameworks • System telemetry & monitoring • Code quality standards • Performance profiling • GDB • Agile methodologies

Additional Skills: Cloud architecture • Docker/Kubernetes • Linux • Git/SVN • High-frequency systems • Low-latency optimization

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PROFESSIONAL EXPERIENCE

Adobe Systems, Bengaluru — Computer Scientist | MAY 2022 – PRESENT

• **CICP Performance Optimization** — Architected end-to-end service optimization achieving 50% performance improvement through strategic caching and multithreading patterns; served millions of concurrent users with mission-critical latency requirements; owned design, implementation, testing, and deployment.

• **Express-Acrobat Backend Architecture** — Designed complete service-oriented backend for PDF editing workflows; architected URN-based content management system with cloud storage integration; implemented IndexedDB caching layer for client-side performance; established CI/CD pipeline and comprehensive test coverage; collaborated cross-functionally with Product Management on feature prioritization and A/B testing strategy.

• **PDF Export Rendering System** — Architected high-fidelity export subsystem with optimized rendering pipeline; designed data structure and algorithm improvements handling complex document transformations at scale; ensured enterprise-grade stability for millions of users.

• **System Reliability & Performance Engineering** — Diagnosed and resolved critical CPU/GPU bottlenecks in AGM using profiling tools; applied multithreading best practices eliminating threading issues; optimized CICP encoding pipeline through algorithm optimization and data structure selection.

• **Technical Leadership & Mentorship** — Led C++ Bootcamp for university graduate engineers establishing foundational expertise; conducted internal knowledge transfer establishing team excellence in Express platform architecture; mentored junior engineers on system design and OOP principles.

Cadence Design Systems, Noida — Software Engineer II | DECEMBER 2020 – APRIL 2022

• **FPGA Simulation Backend** — Designed and implemented Save/Restore service for runtime state capture enabling efficient simulation workflows; architected multiple optimization schemes (buffer-based, directory-based) balancing performance and memory usage; delivered production-grade system with comprehensive test coverage.

• **Memory Optimization & Data Structure Design** — Architected efficient state management subsystem reducing memory footprint by 35%; applied advanced data structure and algorithm techniques; designed for both performance and code quality.

• **Object-Oriented Architecture** — Implemented sophisticated design patterns (Factory, Observer, Strategy) solving complex system requirements; ensured code modularity and maintainability in high-performance simulator.

Azcom Technology, Gurgaon — Software Engineer | JUNE 2018 – NOVEMBER 2020

• **NB-IoT Protocol Stack Architecture** — Led team of 3 engineers designing and implementing control plane protocol stack for Linux-based networking systems; architected modular, scalable service architecture; delivered production-grade system with enterprise reliability serving telecommunications infrastructure.

• **Dynamic System Configuration** — Designed System Lock/Unlock service using runtime polymorphism and state management patterns enabling real-time parameter modification without system restart; ensured system stability during configuration transitions.

• **System Diagnostics & Telemetry** — Developed memory leak detection framework leveraging STL containers and signal handlers; implemented runtime profiling and monitoring enabling proactive incident detection; significantly reduced production issues.

• **Performance Optimization** — Optimized critical code paths using algorithm selection and data structure design; reduced CPU cycles and improved system throughput; applied profiling tools to identify and eliminate bottlenecks.

TECHNICAL ACHIEVEMENTS

Competitive Excellence & Technical Depth:
• Top 0.4% Rank — TCS Codevita 2017 (200K+ participants across Rounds 1 & 2)
• 1746 Maximum Rating — HackerRank
• 100+ Problems Solved — LeetCode (advanced algorithms and data structures)
• 7 Elite-level IIT Certifications in advanced computer science core subjects

Recognition & Impact:
✓ Company hackathon winner (GenAI-powered Illustrated Novels project)
✓ Architected and led multiple large-scale system initiatives

✓ Established organizational expertise in high-performance backend systems
✓ Consistently delivered mission-critical features ahead of schedule with high quality

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TECHNICAL STACK

Languages: C++ (advanced — modern C++11/17/20, STL, multithreading) | Python | Java | TypeScript | JavaScript | Bash

Tools & Platforms: GDB | Git/SVN | Linux | Docker | Kubernetes | CI/CD tools | Profiling & debugging tools | Qt | Wireshark | Express.js | Flask

Databases: MySQL | MongoDB | Cloud storage systems

Specializations: Large-scale distributed systems • Object-oriented design • Service-oriented architecture • High-frequency systems • Low-latency optimization • Multithreading & concurrency • System performance engineering • Code quality & testing

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EDUCATION

B.Tech in Electronics & Communication Engineering
Netaji Subhas University of Technology, East Campus, New Delhi(2018) | 79.63%

Key Achievements:
• AISSCE (XII) – 2013, 89% (School Topper: CS & English)
• AISSE (X) – 2011, 9.8/10 CGPA

PROJECTS & INNOVATIONS

Illustrated Novels (GenAI) — Sep-Oct 2024
Built an automated illustration generation system using Groq AI and Firefly API; built end-to-end backend service with Python, TypeScript, and RESTful APIs; won company-wide hackathon recognition.

Mockup Placement (Computer Vision) — Jul-Sep 2023
Designed intelligent frame detection and placement model using YOLOv3; implemented Python backend with Google Colab for model training and inference.